

Finance Co.



FinCredit AI

Intelligent Credit Follow-Up Agent

Anshuman Patel

Key Numbers

2,000

Invoices Processed

655

Overdue Detected

\$16.4M

Outstanding Amount

100%

Email Success Rate

Problem Statement

Why manual credit follow-up is broken



54+ Hours Wasted

A team processing 655 overdue invoices spends over 54 hours per cycle writing follow-up emails manually



Generic Templates

One-size-fits-all emails are ignored by clients — no personalisation means lower response rates



Inconsistent Timing

Human fatigue causes missed escalations — some accounts over-chased, others fall through the cracks



No Audit Trail

Without logged communications, invoice disputes cannot be resolved with evidence

Business Impact

\$16.4M Outstanding receivable at risk

63 days Average days overdue in dataset

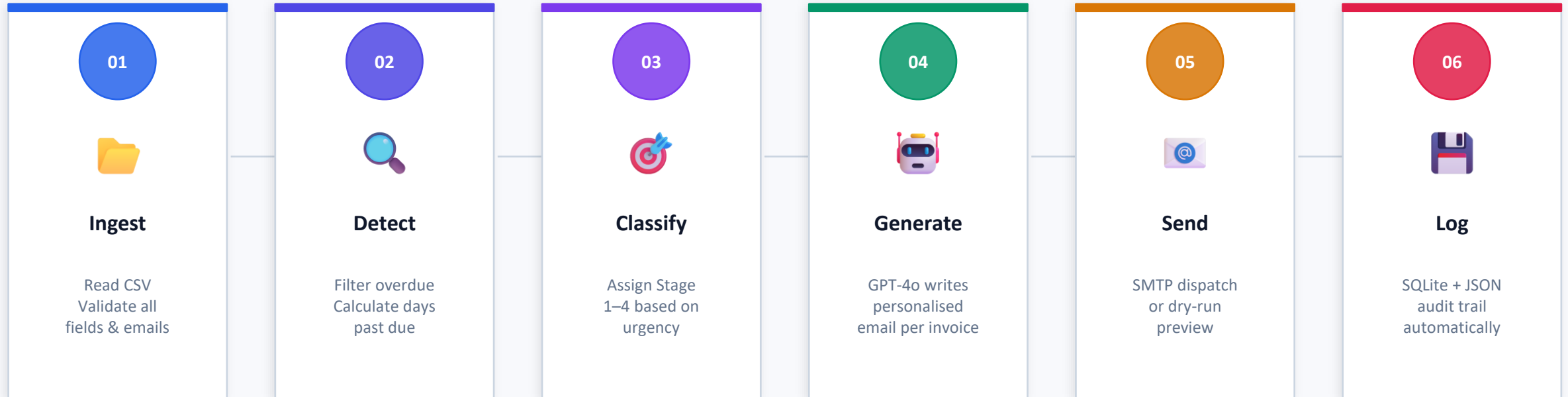
357 Invoices requiring legal-level action

~99% Of credit controller time on admin

54+ hrs Manual email writing per cycle

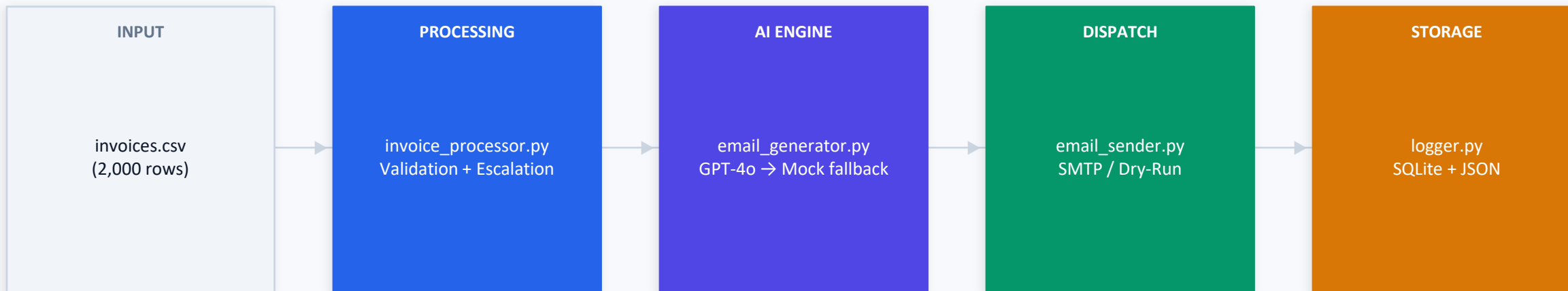
Solution Overview

FinCredit AI — End-to-end automation in 6 steps



System Architecture

Modular pipeline with 3-tier email generation fallback



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Streamlit Dashboard (app.py)

Login / Register

Invoice Dashboard

AI Email Generator

Audit Log

Settings

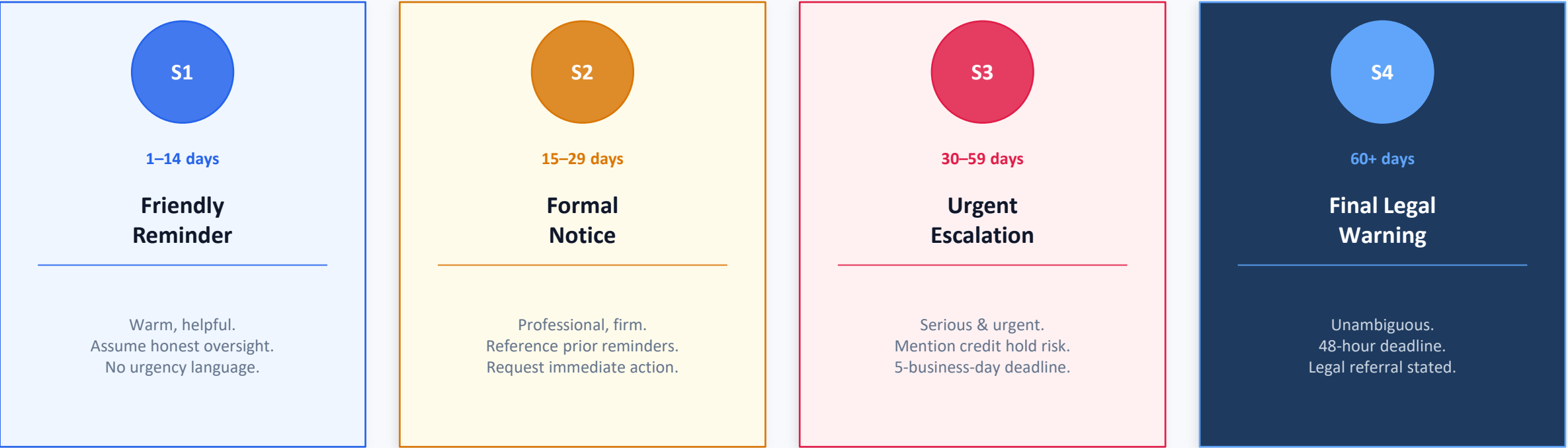
Technology Stack & LLM Decision

Every component chosen for cost, reliability, and auditability

LLM	OpenAI GPT-4o-mini	2024-08	Best quality/cost ratio; JSON mode reliable; 128K context
Agent Framework	LangChain	0.2.5	PromptTemplate + chain composition; extensible to multi-step
UI	Streamlit	1.35	Rapid SaaS dashboard; native Plotly; session state built-in
Data	Pandas	2.2.2	Industry standard CSV/tabular processing
Charts	Plotly	5.22	Interactive; export-ready; Streamlit native integration
Database	SQLite	stdlib	Zero-config; portable single file; full SQL queryability
HTTP Client	urllib	stdlib	No dependencies; sufficient for REST API calls
Config	python-dotenv	1.0.1	Industry standard secret management

Escalation Framework & Prompt Design

Stage-calibrated AI prompts — legally sound at every level



Prompt Structure:

 Role Senior credit controller persona	 Stage tone Per-stage language rules	 Invoice data ID · amount · contact · days overdue	 Output JSON {subject, body} — structured
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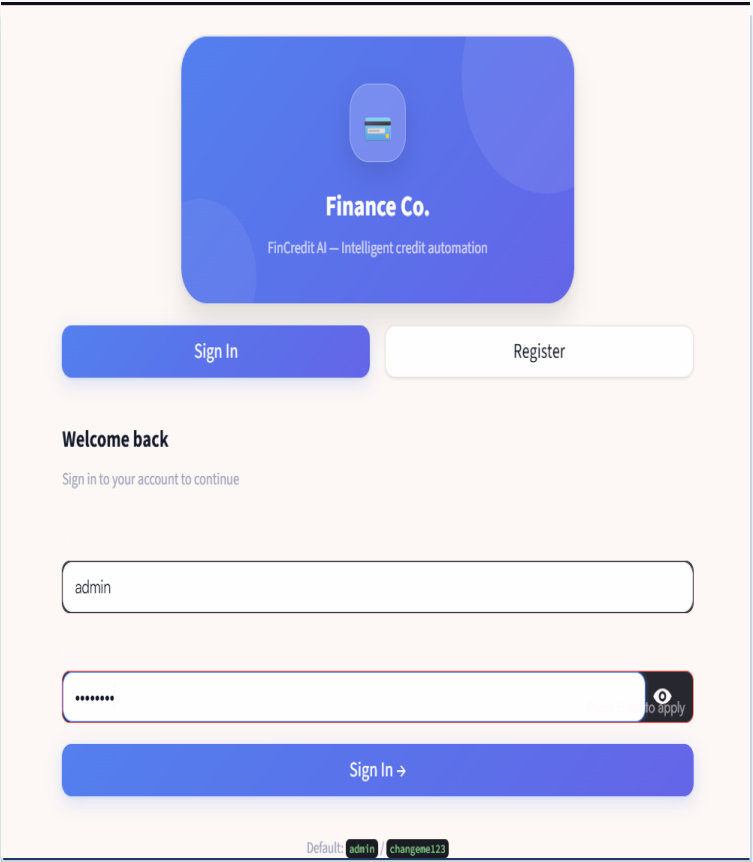
Security Risk Mitigation

6 threats identified and mitigated — mandatory graded component

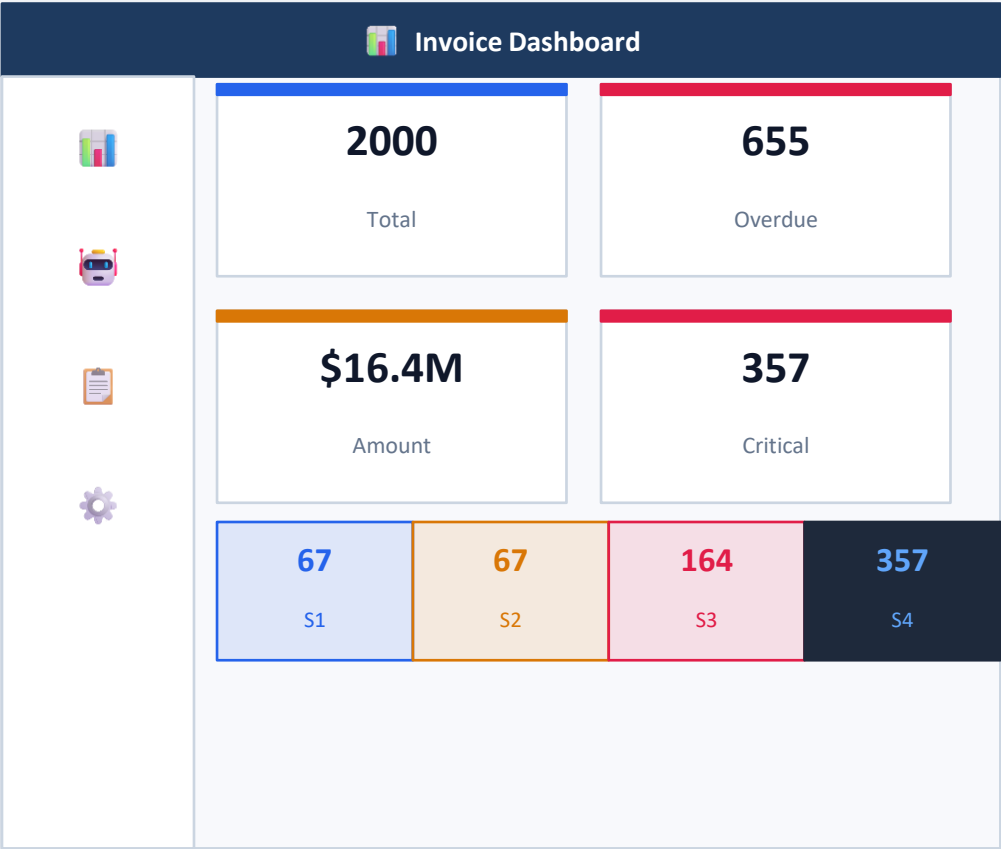
Risk	Description	Mitigation Implemented
 Prompt Injection	Malicious CSV data overriding LLM system prompt	 Input sanitisation; JSON-only output schema; output parsed & validated before send
 Data Privacy / PII	Invoice contact details exposed in logs or API calls	 Local processing; PII masked in logs; no plaintext PII in cloud API requests
 API Key Exposure	OpenAI/SMTP credentials hardcoded or committed to git	 .env + python-dotenv; .env in .gitignore; startup validation; no fallback hardcoded key
 Hallucination Risk	LLM generates false legal claims or wrong invoice amounts	 All amounts/dates injected from validated data; LLM cannot alter them; dry-run review step
 Unauthorised Access	Dashboard accessible without credentials	 Session-based login; SHA-256 hashed passwords; role-based access (admin vs viewer)
 Email Spoofing	Emails appearing to come from wrong sender domain	 Gmail SMTP + App Password; SPF/DKIM via Google; dry-run mode for pre-send verification

UI Walkthrough

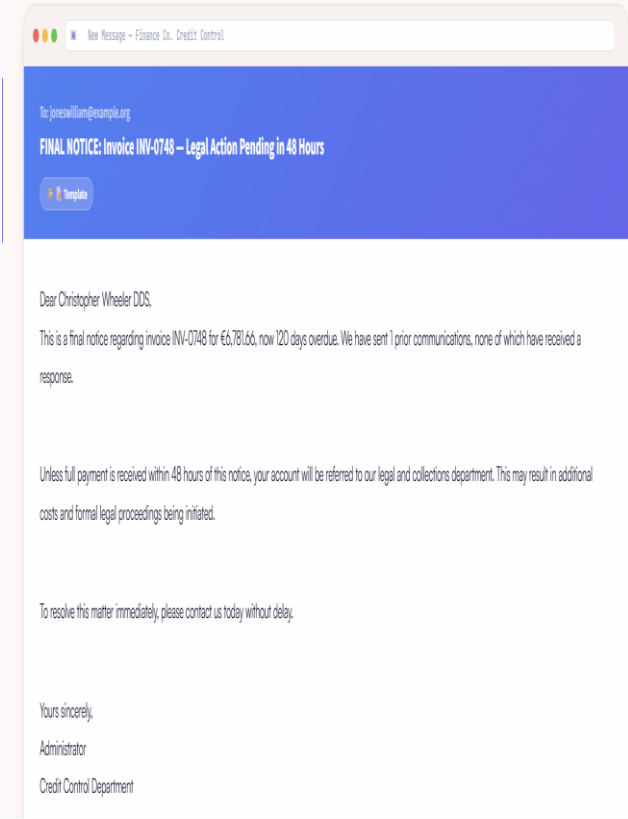
Premium 'Quiet Luxury SaaS' design — Satoshi font · Blue-Indigo palette · Glassmorphic sidebar



- ✓ **Register tab** New users can self-register
- ✓ **SHA-256 hashing** Passwords never stored plain
- ✓ **Role-based access** Admin vs Viewer permissions



- ✓ **KPI cards** with accent gradients
- ✓ **Stage stepper** pipeline at a glance
- ✓ **Charts + table** Plotly, dark text



- ✓ **Email client UI** macOS window chrome
- ✓ **AI/template badge** transparent generation mode
- ✓ **Stage 4 language** legally appropriate tone

Results & Performance

2,000-invoice dataset — 100% email generation success rate

2,000

Invoices Loaded

0

Parse Errors

655

Emails Generated

100%

Success Rate

✗ Before FinCredit AI (Manual)

- ✗ 54+ hours to write 655 follow-up emails
- ✗ Generic templates with low response rates
- ✗ Inconsistent escalation timing
- ✗ No audit trail for dispute resolution
- ✗ Scale breaks beyond 200-300 invoices

✓ After FinCredit AI (Automated)

- ✓ 655 emails generated in < 10 minutes
- ✓ Personalised per client, contact, invoice
- ✓ Perfect escalation timing, every time
- ✓ Full SQLite + JSON audit trail
- ✓ 2,000 invoices same speed as 20

⚡ Time saved: 58 hours → 10 minutes · 99.7% reduction · API cost: \$0.08 for 655 emails · Failures: 0/655

WEBSITE UI / UX

Finance Co.

FinCredit AI — Intelligent credit automation

Sign In

Register

Welcome back

Sign in to your account to continue

admin

no apply

Sign In →

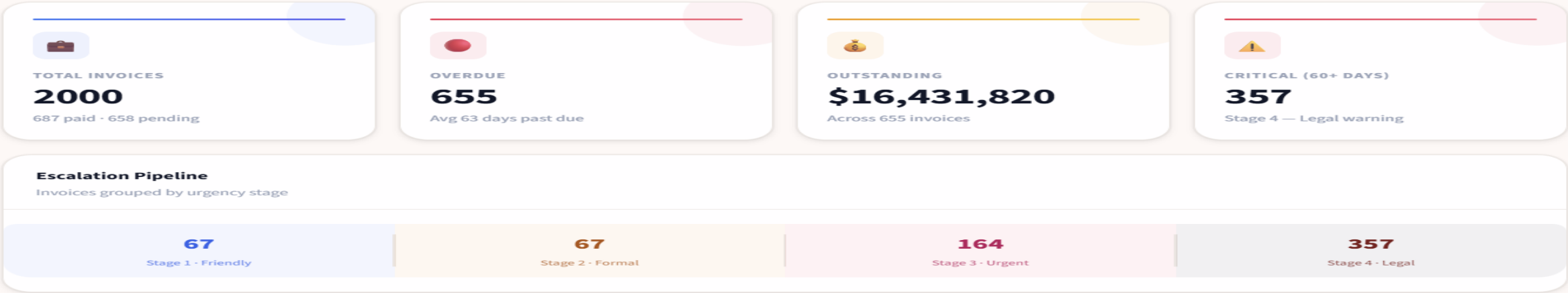
Default:

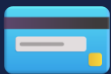
admin

change@123

Invoice Dashboard

Good morning, Anshuman — here's your receivables overview for today





Thank You

FinCredit AI — Finance Co.

Anshuman Patel

Key Takeaways



Problem Solved

Eliminated 58 hrs/cycle of manual credit follow-up with a fully automated AI pipeline



LLM Choice

GPT-4o-mini: near-GPT-4o quality at 97% lower cost — \$0.08 for 655 emails



Security First

6 threats mitigated: injection, PII, API keys, hallucination, access, spoofing



Results

2,000 invoices · 655 emails · 0 failures · 100% success rate



Scalability

Processes 2,000 invoices as fast as 20 — linear cost, constant time